

PRAYCG Master Comprehensive Analysis Suite

Detailed module guide for v1.5.5: A-MRED, MRED-Peak/Resolution, NUPI, and offline interpretation

Public-facing technical guide - exploratory methods architecture, not endpoint certification

How to read this guide

The suite is intentionally overcomplete. For public and confirmatory interpretation, use the boxed primary path first: Timing/QC + StimulusFingerprint/CET-R + artifact/confound gates -> A-MRED / MRED-Peak-Resolution. NIP, TTI, and NUPI are secondary summaries. KHT-topo, NAST, EET, MRED-ITP, OCM/RSM/CVB/SquintProxy, and LSO are exploratory/convergence modules.

1. Module tier map

Tier	Modules	Role
BOXED PRIMARY PATH	Run metadata, timing/QC, ALS, StimulusFingerprint/CET-R, artifact/confound gates, A-MRED, MRED-Peak/Resolution	Determines whether an anchor is interpretable and whether Target-specific recognition + integration/recovery is present.
SECONDARY SUMMARIES	NIP/BIT/CII/IAQ, TTI, NUPI, Baseline1-vs-Baseline2	Summarizes immersion proxy, reception/extraction, update polarity, and after-state recovery.
EXPLORATORY / CONVERGENCE	KHT-topo, NAST, EET, MRED-ITP/ACG/OCU, OCM025/RSM/CVB/SquintProxy, LSO/SPM	Mechanism-proxy, burden, echo, ocular, and task-diagnostic layers. Useful for interpretation and falsification; not primary proof.
VISUAL / REPORTING	MasterSync Visualizer and Offline Interpretive Report Generator	Makes synchronized review and local plain-English interpretation possible.

2. Core file flow

Stage	Generated/used files	Why it matters
MediaPrep	Target/Override/Control MP4, cue_schedule JSON/CSV, draft/locked anchors, qc/stimulus regressors.	Defines the physical stimulus and external input u(t).
Runner	events.csv/json, run_config_media_selection.json, predeclared anchors record, self-report JSON, confound JSON, XDF.	Records what actually happened and what was loaded.
Master Suite	feature table CSV, module tables, overlays, reports.	Computes physiology and endpoint summaries.
Visualizer	synchronized MP4, features_used.csv, events_used.csv, render_report.json.	Audit/review artifact, not endpoint certification.
Offline interpreter	offline_interpretive_report.md/txt/json.	Rule-based local interpretation for users without live expert review.

1. Run metadata and media identity

Field	Detail
Tier	Primary QC
Purpose	Verifies participant/session labels, protocol version, branch videos, cue schedule, anchor file, hash identity, and configured durations.
Inputs	Runner event log, run_config_media_selection.json, media hashes, cue schedule summary.
Algorithmic logic	Parse setup markers and sidecar JSON. Confirm Target/Override hash match. Record cue count, expected sum, baseline/washout durations, anchor-file status, channel-map status, and PRAYCG version.
Outputs	run_identity_summary.csv, module registry rows, interpretation JSON fields.
Interpretation	A valid run should show the selected media and cue schedule unambiguously. Target and Override should be hash-identical when the task differs only by instruction.
Failure modes	Missing config, Target/Override mismatch, unrecorded anchor file, absent cue schedule, or unknown protocol version downgrades claims.
Claim status	QC / validity gate

2. Stream inventory and timebase reconstruction

Field	Detail
Tier	Primary QC
Purpose	Determines which LSL streams are present and whether physiology can be aligned to event markers.
Inputs	XDF streams, StasisMarkers, OpenBCIStatusMarkers, sample timestamps.
Algorithmic logic	Inventory stream names, sample rates, channel counts, durations, time gaps, and marker intervals. When raw timestamps are unreliable, reconstruct the OpenBCI timebase using sample-index and status markers.
Outputs	stream_inventory_corrected.csv, phase_coverage.csv.
Interpretation	Successful reconstruction supports phase-sliced features even when raw XDF timestamps are imperfect.
Failure modes	Missing EEG stream, major dropout, unsolved timebase inconsistency, or poor phase coverage prevents strong interpretation.
Claim status	QC / validity gate

3. ALS/PT19 display-timing QC

Field	Detail
Tier	Primary QC
Purpose	Checks whether physical screen transitions were detected at Control, Target, and Override starts.
Inputs	ALS_PT19_Timing stream or OpenBCI analog AUX, branch start markers, cue/media timing manifest.
Algorithmic logic	Detect black-white-black or timing-square pulses around

	branch-start markers. Estimate onset offset, duration, clipping, and pulse presence.
Outputs	als_start_pulse_qc.csv, ALS timing summary.
Interpretation	A clean pass strengthens display-onset alignment. ALS belongs to external input u(t), not biology.
Failure modes	Missed or ambiguous pulses downgrade anchor timing and endpoint claim grade.
Claim status	QC / timing gate

4. Channel map and ROI audit

Field	Detail
Tier	Primary QC
Purpose	Checks whether channel-to-electrode mapping is confirmed before ROI-based interpretation.
Inputs	channel_map.csv, event markers, run configuration.
Algorithmic logic	Parse PRAYCG16 channel map markers and confidence flags. Mark ROI use as allowed, cautioned, or blocked.
Outputs	channel_map_summary.csv or module registry field.
Interpretation	LOCKED channel map allows frontal/parietal/occipital/temporal ROI labels to be used cautiously.
Failure modes	Unconfirmed channel map blocks ROI-specific claims even if global features can still be analyzed.
Claim status	QC / ROI gate

5. Feature table generation

Field	Detail
Tier	Primary engine
Purpose	Creates the time-resolved table used by all downstream modules.
Inputs	Cleaned/filtered EEG, event markers, HR/RR, respiration, ALS, artifact channels, StimulusFingerprint regressors where available.
Algorithmic logic	Slice by phase, compute sliding-window features, spectral bands, ROI summaries, artifact sentinels, autonomic summaries, and condition labels. Typically writes 1 s rows; OCM025 uses 0.25 s cue windows.
Outputs	* time_resolved_feature_frame.csv.
Interpretation	This is the canonical physiology feature table. Visualizer *_features_used.csv is only the subset used for a render.
Failure modes	Missing feature table means most downstream modules cannot run.
Claim status	Feature extraction basis

6. ArtifactScore

Field	Detail
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Tier	Primary gate
Purpose	Quantifies windows where EEG features may be contaminated by blink, jaw, motion, flatline, or high-frequency noise.
Inputs	EEG channels, Fp1/Fp2/frontal channels, high-frequency sentinel bands, peak-to-peak amplitude.
Algorithmic logic	Combine normalized peak-to-peak, line/high-frequency sentinel activity, ocular/frontal outlier features, and dropout/flatline checks. Penalize or veto contaminated windows.
Outputs	artifact_score columns, artifact tables, module veto flags.
Interpretation	ArtifactScore has veto power. A gamma/theta event that is artifact-driven is not an endpoint.
Failure modes	High artifact in candidate windows converts biological interpretation into artifact/confound caution.
Claim status	Artifact gate

7. StimulusFingerprint v1.8

Field	Detail
Tier	Primary guardrail
Purpose	Extracts exogenous stimulus regressors from the final rendered media.
Inputs	Control/Target/Override MP4s, cue schedule JSON, locked anchor JSON if available.
Algorithmic logic	Compute luminance, visual change, histogram delta, cut events, flash proxy, audio RMS/envelope derivative, cue trains, cue phase, ALS/start-pulse regressors, and anchor stimulus vectors.
Outputs	qc/stimulus_exogenous_regressor_frame_all_conditions.csv, cet_regressors_all_conditions.csv, stimulus_rhythm_summary_all_conditions.csv.
Interpretation	This lets CET-R test whether physiology is explained by stimulus rhythm, cue rhythm, luminance, cuts, motion, or audio envelope.
Failure modes	Missing Target/Override regressors makes CET-R partial. Human QC is still required for control audio unintelligibility.
Claim status	Stimulus-side control

8. CET - Cinematic Entrainment Tracking

Field	Detail
Tier	Primary guardrail / exploratory
Purpose	Estimates how much EEG/NIP-like features track external film/cue/audio/visual rhythm.
Inputs	StimulusFingerprint regressors and feature table.
Algorithmic logic	Fit stimulus-side regressors to physiology features within each branch. Evaluate train and blocked-cross-validation R2.
Outputs	cet_residualization_model_summary.csv, cet_cue_tracking_summary.csv.

Interpretation	High generalizable R2 means apparent effects may be stimulus tracking. Low/negative blocked-CV suggests stimulus regressors do not explain the module pattern.
Failure modes	Weak regressors, missing audio/video features, or overfit models prevent strong conclusions.
Claim status	Confound/residualization guardrail

9. CET-R - Residualized meaning metrics

Field	Detail
Tier	Primary guardrail
Purpose	Re-tests NIP/MRED/TTI after regressing out external media rhythms.
Inputs	Feature table, CII/NIP/MRED metrics, StimulusFingerprint regressors.
Algorithmic logic	Model features as functions of exogenous regressors, compute residuals, then recompute anchor integrals/contrasts.
Outputs	cet_residualized_cii_anchor_integrals.csv, residualized summaries.
Interpretation	If Target remains greater than Control/Override after CET-R, the result is less likely to be reducible to raw stimulus rhythm.
Failure modes	If effects vanish after residualization, meaning-specific interpretation weakens.
Claim status	Primary guardrail, not proof

10. GammaScalpel

Field	Detail
Tier	Feature extraction / secondary
Purpose	Separates lower-gamma candidate activity into narrower bands and context labels instead of treating gamma as meaning.
Inputs	EEG feature windows, ROI labels, artifact scores, condition labels.
Algorithmic logic	Subdivide lower gamma, e.g. 30-35, 35-40, 40-45 Hz, compare Target/Control/Override, separate power from phase-locking, apply artifact and visual/task penalties.
Outputs	gammascalpel_microband_phase_summary.csv, MeaningGamma/TaskGamma components.
Interpretation	Gamma is an artifact-sensitive candidate work-signal proxy, not a biomarker. It contributes to MR only if gates survive.
Failure modes	EMG/ocular/artifact, cue/task load, or visual features can explain gamma-like changes.
Claim status	Feature extractor / artifact-sensitive

11. Temporal Semantic Proxy (TSP)

Field	Detail
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Tier	Feature extraction / secondary
Purpose	A composite proxy for scene-timed semantic recognition pressure.
Inputs	MeaningGamma proxy, taper/peak features, artifact penalties, condition labels, anchor windows.
Algorithmic logic	Score temporally localized recognition-like events using meaning-weighted gamma/TSP components, taper detection, and artifact-corrected normalization.
Outputs	TSP columns, TSP event candidates, visual overlays.
Interpretation	TSP marks candidate semantic strike timing. It is not proof of comprehension or memory.
Failure modes	TSP can be driven by visual salience, audio, task, or artifact if not controlled.
Claim status	Meaning-recognition proxy

12. CandidateLocal KHT-topo

Field	Detail
Tier	Exploratory convergence
Purpose	Tests local reciprocal coupling between observed work-signal features and inferred network-state topology proxies.
Inputs	Feature table, candidate events, theta/meaning/TSP windows.
Algorithmic logic	Fit event-local reciprocal loop models using recognition features and delayed integration/topology features; compute K_HT-topo and percentiles.
Outputs	candidate_local_kht_topo_mred_event_table.csv, kht visual overlays.
Interpretation	K_HT-topo is a human-scale coupling proxy. It is not K_OSM and not cellular proof.
Failure modes	High K without delayed theta/carryover or condition specificity is insufficient.
Claim status	Exploratory mechanism-proxy

13. MRED - Meaning Recognition / Encoding Dissociation

Field	Detail
Tier	Primary conceptual endpoint
Purpose	Separates meaning recognition (MR) from encoding/integration (ENC).
Inputs	TSP, MeaningGamma, KHT-topo, theta handoff, topology/afterstate proxies, artifact and condition labels.
Algorithmic logic	Compute MR and ENC scores at anchors/candidates. Classify quadrants such as MR_HIGH_ENC_HIGH or MR_HIGH_ENC_LOW.
Outputs	mred_event_table.csv, candidate_local_kht_topo_mred_event_table.csv.
Interpretation	MRED is the conceptual core: recognition can appear without clear integration, and integration-like changes may be nonsemantic unless Target-specific.
Failure modes	No Control/Override specificity, high artifact, or no delayed window weakens classification.

Claim status	Primary conceptual endpoint
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14. A-MRED - Anchor-Locked MRED

Field	Detail
Tier	Boxed primary path
Purpose	Compresses the suite into a cleaner primary endpoint for future studies.
Inputs	Locked anchor JSON, feature table, MRED table, Control/Target/Override contrasts, QC gates, CET-R where available.
Algorithmic logic	For each predeclared anchor, require Target MR and ENC above thresholds and greater than Control and Override, with QC pass.
Outputs	amred_anchor_endpoint_table.csv, amred_primary_endpoint_summary.csv, amred_visual_overlay.csv.
Interpretation	A-MRED pass is the strongest current endpoint category. Strict use requires frame-verified locked anchors and clean timing.
Failure modes	Estimated anchors, weak ALS, missing Control/Override, or timing sensitivity produce pilot-positive, not confirmatory, results.
Claim status	Primary endpoint

15. MRED-Peak

Field	Detail
Tier	Boxed primary path / endpoint subtype
Purpose	Identifies acute anchor-locked meaning shock or recognition-plus-integration events.
Inputs	A-MRED, BIT, MRED scores, CII margins, KHT/MR/ENC tables.
Algorithmic logic	Combine target MR, target ENC, strict gates, CII margins, and QC flags into a peak score and gate label.
Outputs	mred_peak_resolution_anchor_table.csv, top_mred_peak_candidates.csv.
Interpretation	Best for Contact-style acute recognition events. A strict peak pass still requires anchor/timing confidence.
Failure modes	Peak candidates without strict lock stay exploratory.
Claim status	Primary endpoint subtype

16. MRED-Resolution

Field	Detail
Tier	Secondary/endpoint subtype
Purpose	Identifies delayed reflective or regulatory recovery rather than a single acute peak.
Inputs	Baseline2 metrics, NUPI/RDI, EET, self-report echo,

	MRED/NIP scores.
Algorithmic logic	Combine post-target recovery, TargetEchoedDuringBaseline2, EET afterstate echo, and RDI with anchor-level context.
Outputs	mred_peak_resolution_anchor_table.csv, top_mred_resolution_candidates.csv.
Interpretation	Best for Field-of-Dreams-style closure/recovery. It is a profile, not proof of healing.
Failure modes	No Baseline2 means true resolution cannot be graded; proxy-only results are flagged.
Claim status	Secondary endpoint subtype

17. NIP - Narrative Immersion Proxy

Field	Detail
Tier	Secondary
Purpose	Translates attention + resonance into macroscopic PRAYCG proxies.
Inputs	MeaningGamma/TSP/NAST for semantic attention; theta/KHT/topology/API for integration; artifact/confounds.
Algorithmic logic	Compute A_sem and R_int components; multiply/compose into immersion density.
Outputs	nip_component_timeseries.csv.
Interpretation	NIP estimates attention-plus-integration immersion proxy, not dopamine/oxytocin or literal immersion chemistry.
Failure modes	High sensory drive or artifact can mimic attention; high recognition without integration should not be called full immersion.
Claim status	Secondary summary

18. BIT - Bivariate Immersion Threshold

Field	Detail
Tier	Secondary
Purpose	Ensures immersion is not a mere average of attention and resonance.
Inputs	NIP components, MRED quadrants, artifact and condition gates.
Algorithmic logic	Boolean gate requiring semantic attention and integration/resonance to exceed thresholds simultaneously, with condition specificity.
Outputs	bit_event_table.csv.
Interpretation	BIT pass approximates a strict MR_HIGH_ENC_HIGH immersion candidate.
Failure modes	High attention but low integration fails BIT by design.
Claim status	Secondary event gate

19. CII - Continuous Immersion Index

Field	Detail
Tier	Secondary
Purpose	Scores scene-level immersion density over anchor windows.
Inputs	NIP time series and anchor windows.
Algorithmic logic	Integrate/average positive semantic-attention multiplied by integration proxy across a window, duration-normalized.
Outputs	cii_anchor_integrals.csv.
Interpretation	Useful for sustained scenes, not just spikes. Longer windows must be normalized.
Failure modes	Can be inflated by duration, sensory rhythm, or artifacts if CET-R and QC are missing.
Claim status	Secondary summary

20. IAQ / ITQ - Immersion Attenuation

Field	Detail
Tier	Secondary
Purpose	Estimates how much Override attenuates Target immersion proxy.
Inputs	Target and Override CII/NIP values for matched anchors.
Algorithmic logic	Compute Target-minus-Override difference or ratio-bounded attenuation.
Outputs	iaq_target_override_table.csv.
Interpretation	Positive IAQ means Target preserved more immersion proxy than Override. It is not a moral theft score.
Failure modes	If Override equals/exceeds Target, reception/extraction claim weakens or story breaks through task.
Claim status	Secondary contrast

21. TTI - Thermodynamic Theft Index

Field	Detail
Tier	Secondary
Purpose	Summarizes reception-vs-extraction tradeoff.
Inputs	MR, ENC, API-A/autonomic summaries, task/extraction metrics, artifact/visual penalties.
Algorithmic logic	Weighted contrast: Target meaning/integration/regulation minus Override extraction/task load, penalized for artifacts and visual differences.
Outputs	tti_global_summary.csv, tti_anchor_deltas.csv, tti_visual_overlay.csv.
Interpretation	Positive TTI means Target preserved more receptive meaning/integration while Override carried more extractive load.
Failure modes	Not literal thermodynamics, morality, or clinical metric. Artifact or task confounds can dominate.
Claim status	Secondary summary

22. NUPI - Narrative Update Polarity Index

Field	Detail
Tier	Secondary
Purpose	Classifies update profile as accommodative load versus resolute recovery.
Inputs	ALI components, RDI components, Baseline2, NUPI anchors, self-report echo, EET.
Algorithmic logic	Compute ALI as semantic/update-load profile; compute RDI as regulatory-dividend/afterstate profile; NUPI = RDI - ALI.
Outputs	nupi_run_summary.csv, nupi_anchor_polarity_table.csv, nupi_visual_overlay.csv.
Interpretation	Distinguishes high-load-with-recovery from resolute-recovery profiles. Does not measure heat/ATP/clinical healing.
Failure modes	No Baseline2 or no afterstate physiology prevents reliable polarity grading.
Claim status	Secondary interpretation

23. NAST - Narrative Absorption State Transition

Field	Detail
Tier	Exploratory / transition
Purpose	Detects branch-entry state transition into narrative absorption or task stance.
Inputs	Late washout windows, early branch windows, alpha, MeaningGamma, TSP, TaskGamma, theta.
Algorithmic logic	Compare pre-branch and early-branch features; quantify meaning/alpha/task transitions.
Outputs	nast_phase_transition_table.csv.
Interpretation	Useful for onset dynamics. Target may show meaning/TSP increase; Override may show task-gamma increase.
Failure modes	Alpha is eye/attention sensitive; NAST alone is not a meaning endpoint.
Claim status	Exploratory transition layer

24. Topo-OSM NetworkState

Field	Detail
Tier	Exploratory interpretation
Purpose	Represents human-scale latent state as network-state topology proxy.
Inputs	Theta, gamma, alpha, TSP, API-A, respiration, artifact, branch summaries.
Algorithmic logic	Build phase/branch vectors and transition distances rather than cellular hidden-Y claims.
Outputs	topo_osm_phase_state_vectors.csv, topo_osm_visual_overlay.csv.
Interpretation	Human-scale Topo-OSM only: state geometry, not microtubules or biophotons.
Failure modes	Too few channels, unconfirmed channel map, or artifact can make topology labels unreliable.

Claim status	Exploratory interpretation
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25. EET - Endogenous Echo Tracking

Field	Detail
Tier	Exploratory convergence
Purpose	Tests whether scene-anchor state vectors resemble later washout/Baseline2 states.
Inputs	Anchor feature vectors, washout/Baseline2 feature vectors, NIP/MRED/EET tables.
Algorithmic logic	Compute cosine/similarity or vector resemblance between anchor state and later afterstate windows.
Outputs	eet_endogenous_echo_tracking.csv.
Interpretation	Suggests afterstate resemblance. It is not replay proof or memory proof.
Failure modes	Similarity may reflect baseline drift, common arousal, artifact, or generic state persistence.
Claim status	Exploratory afterstate layer

26. MRED-ITP - Information-Thermodynamic Proxy

Field	Detail
Tier	Exploratory convergence
Purpose	Adds complexity and ocular-release proxies around MRED events.
Inputs	LZ-style complexity time series, blink/Fp1 proxy, MRED anchors, artifact.
Algorithmic logic	Compute complexity strike/settlement and blink suppression/release metrics, then summarize convergence with MRED.
Outputs	lz_complexity_timeseries.csv, acg_event_table.csv, blink_event_table.csv, ocu_event_table.csv, mred_itp_anchor_summary.csv.
Interpretation	A convergence layer only. Complexity is not thermodynamic entropy; blink release is not memory encoding proof.
Failure modes	High artifact, lack of EOG/eye tracking, or modality-specific complexity effects limit interpretation.
Claim status	Exploratory

27. ACG - Algorithmic Complexity Gate

Field	Detail
Tier	Exploratory
Purpose	Looks for complexity perturbation and settlement around anchors.
Inputs	Feature-state or raw EEG symbolic complexity windows.
Algorithmic logic	Compare pre, peak, and post LZ-style complexity: strike = peak-pre; settle = post-peak.
Outputs	acg_event_table.csv, lz_complexity_timeseries.csv.

Interpretation	Structured complexity trajectory can support perturbation/settlement interpretation.
Failure modes	Not literal entropy; artifact/noise can raise complexity; drowsiness/flatline can lower it.
Claim status	Exploratory

28. OCU - Ocular-Cognitive Unloading

Field	Detail
Tier	Exploratory / ocular
Purpose	Tracks blink suppression and release around narrative anchors.
Inputs	Fp1/frontal ocular proxy or EOG if available, anchor windows, artifact table.
Algorithmic logic	Detect blink-like events, compute blink rate before/during/after anchor, quantify suppression/release.
Outputs	blink_event_table.csv, ocu_event_table.csv.
Interpretation	Possible event-boundary/attention-release marker. Stronger with EOG/eye tracking.
Failure modes	Fp1 proxy confounds blink with squint, EMG, eye strain, fatigue, or electrode motion.
Claim status	Exploratory ocular layer

29. OCM025 - Override Cue Microstate

Field	Detail
Tier	Override diagnostic
Purpose	Analyzes digit recognition and working-memory updates around number cues at 0.25 s resolution.
Inputs	Cue schedule, raw/feature windows, Override/Target/Control comparisons.
Algorithmic logic	Segment cue epochs into pre-cue, digit recognition, working-memory update, and maintenance windows; compare to nulls.
Outputs	ocm025_rsm_cvb_squint_cue_epoch_table.csv.
Interpretation	Tests whether Override task was physiologically engaged. Not a meaning endpoint.
Failure modes	Poor cue visibility, missed cues, or artifacts can dominate.
Claim status	Override/task diagnostic

30. RSM - Running Sum Microstate Model

Field	Detail
Tier	Override diagnostic
Purpose	Models arithmetic load, compute stall, deadline pressure, and latent guess risk.
Inputs	Cue values, running-sum trajectory, OCM025 features, task self-report.
Algorithmic logic	Compute carry difficulty, high-carry load, hard-9 patterns,

	theta maintenance, closure delay, and latent guess-risk proxies.
Outputs	rsm_correlation_summary.csv, cue tables.
Interpretation	Identifies candidate burden/strategy windows, not proof of private strategy.
Failure modes	Requires accurate cue schedule and task report; cannot prove actual guessing.
Claim status	Override/task diagnostic

31. CVB - Cue Visibility / Legibility Burden

Field	Detail
Tier	Override diagnostic
Purpose	Separates cue perception burden from arithmetic compute burden.
Inputs	Cue schedule, visual contrast/legibility metrics, OCM/RSM features, SquintProxy.
Algorithmic logic	Combine weak digit-recognition, high holding burden, low contrast/blur reports, and visual-search proxies.
Outputs	cue legibility burden columns and candidate tables.
Interpretation	Flags possible hard-to-see cues. Particularly useful for Snack Attack and Her-like issues.
Failure modes	Cannot confirm invisibility without frame-level visibility and/or eye tracking.
Claim status	Covariate/diagnostic

32. SquintProxy

Field	Detail
Tier	Artifact/covariate
Purpose	Uses frontal/Fp1-like artifact as possible visual strain sentinel.
Inputs	Fp1/frontal high-frequency, p2p, blink-like sentinels, self-report.
Algorithmic logic	Score eye-strain/squint candidates separately from blink events and EEG endpoints.
Outputs	squint_proxy candidates within RSM/CVB tables.
Interpretation	Useful as covariate only. EOG/eye tracking is needed for confirmation.
Failure modes	Confounds with blink, EMG, brow tension, electrode movement, general effort.
Claim status	Covariate

33. API-A / Autonomic availability

Field	Detail
Tier	Secondary/covariate
Purpose	Summarizes regulated availability rather than simple arousal.

Inputs	HR, RMSSD/SDNN/pNN50, HR slope, respiration if available.
Algorithmic logic	Weighted z-score composite favoring lower HR, higher HRV, stable/regulated trends, and respiratory context.
Outputs	api_a columns, autonomic summaries.
Interpretation	Helps distinguish receptive regulation from strain, but not a clinical measure.
Failure modes	Requires reliable R-R/ECG; respiration/common drive can explain HRV.
Claim status	Autonomic covariate

34. Baseline 1 vs Baseline 2

Field	Detail
Tier	Secondary afterstate
Purpose	Compares pre-run baseline to final reflection baseline.
Inputs	Baseline phase features, HR/HRV, respiration, NIP/TSP/TaskGamma, artifact.
Algorithmic logic	Compute feature deltas and regulation/reflective-state indicators.
Outputs	baseline1_vs_baseline2_summary.csv.
Interpretation	Supports afterglow/recovery interpretation when Baseline2 is cleaner, more regulated, or semantically active.
Failure modes	Can be contaminated by fatigue, report burden, order effects, or artifact differences.
Claim status	Afterstate summary

35. LSO/SPM Subtitle Override

Field	Detail
Tier	Optional exploratory
Purpose	Analyzes subtitle-only or subtitle-mediated override variants.
Inputs	Subtitle file, bounding boxes, optional gaze, feature table, MRED/NIP outputs.
Algorithmic logic	Compute lexical extraction cost, subtitle phase mapping, semantic timing shift, and subtitle MRED patterns.
Outputs	subtitle_line_event_table.csv, subtitle_phase_shift_table.csv, lso_visual_overlay.csv.
Interpretation	Tests route cost: semantic access through text versus natural audiovisual absorption.
Failure modes	Without gaze, reading timing is estimated only. Audio+subtitle variants test a different condition.
Claim status	Optional protocol variant

36. MasterSync Visualizer

Field	Detail
Tier	Visualization/audit
Purpose	Creates synchronized stimulus/data review videos with

	public/TSC or lab-diagnostic modes.
Inputs	Stimulus MP4, XDF or feature CSV, analysis output folder, overlay tables.
Algorithmic logic	Slice branch by markers, repair timebases when possible, robust-scale panels, show missing-data labels, render event overlays/cards.
Outputs	analysis MP4, features_used.csv, events_used.csv, render_report.json.
Interpretation	Excellent for review and explanation. Does not certify endpoint validity.
Failure modes	Bad feature table, missing panels, or uncorrected event timebase can mislead.
Claim status	Audit visualization

37. Offline Interpretive Report Generator

Field	Detail
Tier	Reporting / usability
Purpose	Generates a local rule-based plain-English report from a Master Suite output folder.
Inputs	Analysis folder tables, optional auto-run MRED-Peak/Resolution.
Algorithmic logic	Detect available tables, summarize QC, A-MRED, MRED-Peak/Resolution, NIP/TTI/NUPI/CET/EET/ITP/OCM/RSM, and boundaries.
Outputs	offline_interpretive_report.md/txt/json.
Interpretation	Helps GitHub/OSF users interpret outputs without uploading data to an AI assistant.
Failure modes	Rule-based, not flexible scientific reasoning. It cannot rescue bad inputs or discover unmodeled confounds.
Claim status	Interpretive report tool

Appendix A. Where major files are generated

File	Generated by	Typical location	Purpose
MRED familiarity CSV	Manual/template input	analysis_inputs/<run> or templates/	Novelty/familiarity/current-life covariates.
MRED scene map CSV	Manual/template input derived from anchor schedule	analysis_inputs/<run> or templates/	Maps anchor IDs to rendered times and predicted MRED role.
Annotation CSV	Anchor/scene-map bridge or report package	analysis_inputs/annotations or report package	Human-readable anchor windows and visual overlay context.
Feature table CSV	Master Comprehensive Suite	<analysis_output>/tables/*time_resolved_feature_frame.csv	Canonical time-resolved physiology feature table.
Visualizer features_used.csv	Visualizer	beside rendered MP4	Subset used for a specific render, not canonical feature table.

Appendix B. Recommended public wording

The suite should be described as an exploratory open-source psychophysiology workbench. The safest public claim is: PRAYCG tests whether narrative meaning has timing, shape, and physiological consequence beyond sensory input and analytic task demand. It does not prove consciousness, OSM biology, clinical efficacy, or metaphysics.